

Printed Circuit Board Design in Computer-aided Design Tool

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Abstract—Printed Circuit Board (PCB) design has traditionally been carried out using Electronic Design Automation (EDA) software tools, which are developed mainly for electrical circuit analysis and for simulating the operation of electronic circuits. Recently, CAD software tools, in addition to their primary purpose, have started offer the electrical circuit analysis with PCB design. The advantage of using CAD tools for PCB design lies in their familiar user interface and integration with existing drawing and modeling skills. This integration allows students / engineers to approach PCB design comprehensively, while from an educational perspective, students can more efficiently acquire new skills without the need to learn an entirely new interface. This paper presents a comparison of PCB design using CAD tool and EDA tool with an emphasis on analysis in the field of schematic capture, components library, PCB layout and 3D visualizations. The examples used for the analysis are based on the teaching material of the ‘Circuit analysis with EDA’ course at the graduate study of Electrical engineering at the Zagreb University of Applied Sciences.

Keywords—PCB design, CAD Fusion 360, EDA Multisim / Ultiboard, Electrical circuit analysis, PCB layout visualization, SPICE model, temperature sweep, stress simulation, cooling simulation

I. INTRODUCTION

The design of Printed Circuit Boards (PCBs), along with the design and analysis of electronic circuits during the design phase, is carried out using software tools generally referred to as Electronic Design Automation (EDA). Significant features of EDA tools are well known, such as a wide component library, a comprehensive set of measuring instruments, circuit analysis in various domains, circuit analysis under different circumstances, netlist creation (a standard data set for PCB modules), and other capabilities [1], [2], [3]. One of the advantages of EDA software is the integration between circuit design phases: schematic capture, circuit simulation, and circuit analysis with PCB rule check and layout design. This integration eliminates the need for manual data migration between these phases, thereby reducing potential errors that could arise from manual translation between different software or software modules [4]. Consequently, integration reduces the time and cost of the final product. Computer-aided design (CAD) tools unify modeling, drawing, designing, rendering, simulation, manufacturing, etc. Moreover, they provide users with insights into assembly behavior under a given set of circumstances [1]. Recently, CAD software tools, in addition to their primary purpose, have started to offer electrical circuit analysis along with PCB design and PCB analysis. This is

important because one tool enables all the necessary analyses—both the circuit and the PCB—without additional data transfer from one tool or tool module to another. In today's market, an electronic product is not only evaluated based on functionality but also on its performance under real-world conditions/circumstances, such as a stress, thermal stress and mechanical event simulation. Furthermore, circuit behavior under specific conditions affects the lifecycle of electronic products and contributes to the e-waste problem [5]. Nowadays, the design of products and systems requires interoperability and integration of software, electrical, electromagnetic, and mechanical characteristics. Consequently, CAD and EDA tools need to work together to enable a unified approach to design [6]. Software integration for such purposes allows engineers to approach PCB design comprehensively, while from an educational perspective, students can combine knowledge acquired in various courses and apply it practically in a specific context. Additionally, students can more efficiently acquire new skills without needing to learn entirely new interfaces for several software tools. From a teaching perspective, the incremental addition of skills in an already used CAD tool (for drafting and 3D modeling), combined with the knowledge acquired in other courses, is a more time-efficient way of transferring specific knowledge. The PCB design process/workflow is well known and consists of the following steps:

- defining the circuit specifications and the conditions under which it will operate,
- schematic diagram design, circuit analysis,
- PCB layout design,
- PCB analysis,
- creating technical product documentation/preparing for production.

This article presents a comparison of PCB design capabilities in a CAD tool compared to an already used EDA tool in the course ‘Circuit Analysis with EDA’ at the graduate study of Electrical Engineering at the Zagreb University of Applied Sciences. The software tools used for comparison are Autodesk *Fusion 360* (CAD) and National Instruments *Multisim* (EDA), and the comparison was carried out as part of a student's research for an upcoming master's thesis. The aim of this article is to find out the possibilities of CAD *Fusion 360* in PCB design and analysis, as a continuation of the research in [1]. Therefore, the following section (Section II) refers to defining circuit specifications and working conditions. Section III deals with the existing component library and creating/importing SPICE models for missing electronic components. Additionally, a circuit simulation with DC analysis is presented. Section IV deals with PCB

design with rules checking and 3D layout. Section V presents PCB analysis—temperature sweep analysis, cooling simulation and stress simulation. Finally, Section VI brings the discussion, conclusion and future steps.

II. DEFINING CIRCUIT SPECIFICATIONS AND WORKING CONDITIONS

Defining the circuit specifications, functionality and the conditions under which it will operate, consequently, components selection and its housing suitable for the assembly requirements, is the first step in mentioned workflow.

To compare CAD with EDA, a simple circuit was chosen: an astable multivibrator with bipolar transistors and two light-emitting diodes (LEDs) indicating the active output state. The schematic diagram for the selected circuit is shown in Fig. 1.

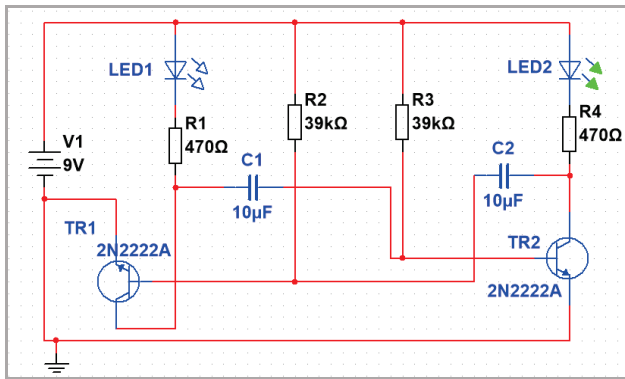


Figure 1. Schematic diagram (Multisim)

Components are selected according to the chosen circuit; the power supply is foreseen via connector (pin header) on the PCB (i.e., the DC power supply is located outside the assembly and is supplied to the assembly by wires). The component housing in both software tools is selected as follows: transistors in TO-18 and TO-18-3 housing, the light-emitting diodes are the radial LED indicator, and the capacitor with resistor in one branch as SMD (Surface Mount Device), while in the other circuit branch, they are in radial housing. The main reason for this is to achieve a better visible thermal overview in software tool, making it easier to observe the difference in component temperatures in related analysis. Furthermore, it is envisaged that the assembly will operate without a housing in a quiet atmosphere without air drafts and (fan) cooling for the same reason. Finally, it is planned to mount the PCB directly to the console via two screws (M3) without additional external loads, which will be used in the *Stress simulation* that will be performed.

III. COMPONENT LIBRARY, COMPONENT SPICE MODEL AND CIRCUIT ANALYSIS

Both tools, *Multisim* and *Fusion 360*, contain component libraries. Each component is described with a SPICE (*Simulation Program with Integrated Circuit Emphasis*) model which is a mathematical model detailing how the component behaves, its functional connections, its type, value, etc. Additionally, each component has at least one PCB model. *Multisim* contains all the necessary components with their own SPICE model.

Unlike *Multisim*, *Fusion 360* doesn't contain all used components in its component library. For example, the transistor and LED have their symbol, footprint and 3D model, but lack a SPICE model. Therefore, it was necessary to import SPICE models for such components. The process is as follows: the component has to be marked in the *Library manager*, right-click on the component in the schematic diagram, *Add Spice Model* option have to be selected and in the *Map* tab the SPICE model has to be copied. In this case, it is taken from *Multisim* (*Select a Component/View model*) and copied to *Fusion 360*, Fig. 2. The same procedure is used for the bipolar transistor. Additionally, SPICE models of some components are offered commercially and free of charge via particular websites (e.g., <https://hr.mouser.com>).

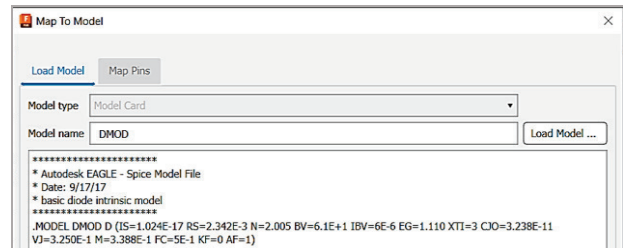


Figure 2. Adding copied component SPICE model from *Multisim* to *Fusion 360*

The second step is to check the circuit functionality. The check was performed by *Operating Point Analysis*. Operating point analysis is used for static analysis of circuits and is also suitable for circuits with nonlinear components such as diodes and transistors. In *Multisim* it can be performed without setting voltage/current probes and it interactively illustrates all node voltages and branch current values including their polarity/direction at all circuit elements, as well as the component power consumption [7]. In *Fusion 360*, only node voltages and the current value in the power supply branch are provided by analysis. For currents in other branches, the ammeters should be placed. The circuit diagram for *Operating point analysis* in *Fusion 360* is shown in Fig. 3.

Obtained results from both software are given in table form and presented in Fig. 4. The results match, although differences appear in the decimal places, as noted in [1] for future analysis. Adjusting the precision does not affect this issue. This analysis is suitable for detecting the maximum current in each circuit branch, which is important information for determining the width of traces (conductors) on a PCB. According to the analysis, the highest current value is in the power supply branch. In Fig. 4 it is marked as $I(V1) = 29.839$ mA in *Multisim*, while in *Fusion 360* the same current is marked as $I(v_v1) = 30.479$ mA. Also, according to the obtained results, the highest power dissipation (obtained in *Multisim*) is on the resistor R1; $P = 105.983$ mW. This information will be used later to select the appropriate component for the PCB. Additionally, if a resistor with lower power dissipation than the results shows is selected, it is necessary to choose another one with the appropriate characteristics from the component library. The same applies to other components. It is worth to be noticed that

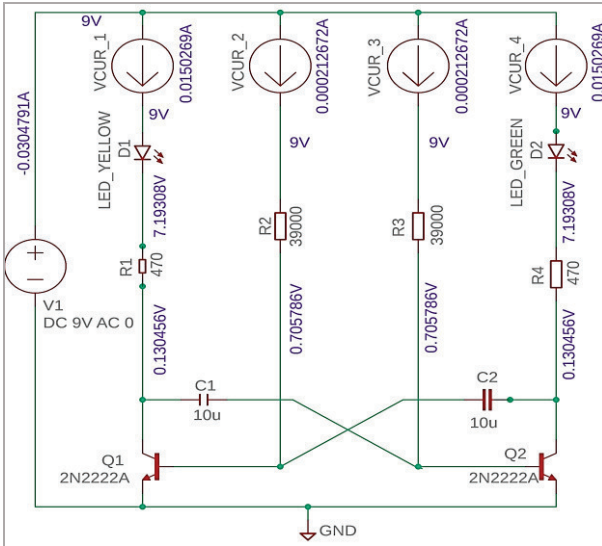


Figure 3. Circuit diagram in *Fusion 360* for Operation Point analysis

MIPRO2025 Operating Point Analysis			Type	Name	Value
Variable	Operating point value				
1 V(9)	70.1.25020 m	1	voltage	v(n_7)	1.304557971259577e-01
2 V(8)	702.47989 m	2	voltage	v(n_1)	7.193078954908001e+00
3 V(6)	129.18676 m	3	voltage	v(n_2)	1.304557971254102e-01
4 V(4)	7.18901	4	voltage	v(n_3)	7.193078954908005e+00
5 V(3)	131.23296 m	5	voltage	v(n_4)	9.000000000000000e+00
6 V(2)	6.89568	6	voltage	v(n_5)	7.057862213228188e-01
7 V(1)	9.00000	7	voltage	v(n_6)	7.057862213228312e-01
8 P(V1)	-268.54995 m	8	voltage	v(n_8)	9.000000000000000e+00
9 P(R4)	97.41577 m	9	voltage	v(n_9)	9.000000000000000e+00
10 P(R3)	1.76588 m	10	voltage	v(n_10)	9.000000000000000e+00
11 P(R2)	1.76535 m	11	voltage	v(n_11)	9.000000000000000e+00
12 P(R1)	105.98342 m	12	current	i(v_v1)	-3.047905986080576e-02
13 P(QTR2)	2.00909 m	13	current	i(v_vcur_1)	1.502685778223167e-02
14 P(QTR1)	2.12012 m	14	current	i(v_vcur_2)	2.126721481712094e-04
15 I(V1)	-29.83888 m	15	current	i(v_vcur_3)	2.126721481712098e-04
16 I(R4)	14.39679 m	16	current	i(v_vcur_4)	1.502685778223167e-02
17 I(R3)	212.78846 u				
18 I(R2)	212.75693 u				
19 I(R1)	15.01655 m				
20 I(QTR2[IE])	-14.60958 m				
21 I(QTR2[IC])	14.39679 m				
22 I(QTR2[IB])	212.78846 u				
23 I(QTR1[IE])	-15.22930 m				
24 I(QTR1[IC])	15.01655 m				
25 I(QTR1[IB])	212.75693 u				

Figure 4. Obtained results for Operating point analysis in *Multisim* and *Fusion 360* (right side)

components are selected in specific housings according to the assembly requirements. Unlike *Multisim*, *Fusion 360* doesn't provide component power consumption, so it needs to be manually calculated for *Cooling simulation*. Despite it is probably not the most important thing, it can be noticed that unlike *Fusion 360*, in *Multisim* it is not possible to mirror the component symbol; symbol for transistor TR (in Fig. 1) and Q1 (in Fig. 3). Although it does not affect the circuit functionality, this results in crossing of conductors in the schematic diagram. For clearer view, it is recommended to avoid cross-overs whenever it is possible, especially in technical product documentation [8]. Finally, based on the obtained results, it can be concluded that *Multisim* is more appropriate for circuit *Operating point analysis*. This analysis could be done with *Fusion 360*, but with some limitations: it doesn't contain SPICE models for all components, so they need to be added manually, and power consumption must be calculated separately for further analysis and component selection.

IV. PCB DESIGN WITH 3D LAYOUT

PCB component model represents housing with Footprint and 3D model with accurate spatial dimensions. Each component has at least one PCB model (one component could be found in many housings; e.g., SMD-A, SMD-B, radial, axial etc.). After checking the functionality of the circuit, it is necessary to transfer the circuit data (*netlist*) from *Multisim* to their PCB module, *Ultiboard*. This can be easily done from main toolbar (*Transfer>Transfer to Ultiboard*). After the data transfer, a Eurocard is opened in both tools. A Eurocard (160x100 mm) is a standardized board format that can be inserted into a standardized subframe or housing. Then follows the manual transfer of components to the Eurocard. Comprehensive guidelines for PCB design (e.g., component position on the board, trace width, avoiding 90-degree angles, the most common hole diameter for component leads passing through holes is 0.8-1 mm...) are given by organization IPC (*Institute for Printed Circuits - Association Connecting Electronics Industries*) and standards they publish (e.g., IPC-2221, IPC-2152...). Standard IEC 60194 is primarily aimed to defining the terminology used in the field of design, production and assembly of PCBs. Both tools use IPC guidelines in PCB design [8], [9], [10], [11]. According to these guidelines, connector for supply is placed along the board edge, as well as LED and transistors. Also, the same guidelines recommend trace widths depending on the maximum current value with 10 °C temperature rise; e.g., for 1 A–10 mil, for 3 A–50 mil, etc., for the most common copper thickness on board – 35 µm. However, some sources offer a *rule of thumb* based on a good practice: 10 mil for currents up to 0.3 A [12], which roughly and broadly account for possible current increases due to temperature rise; the pad diameter around the hole at least 1.8 times hole diameter. Otherwise, a *mil* is 1/1000 of an inch, or 0.0254 mm. In both tools, the measures (trace width, clearance, copper pads...) can be set to *mil* or *mm*.

Default trace width in *Fusion 360* is 6 mil, while in *Ultiboard* is 10 mil. Thus, trace width in *Fusion 360* is 0.1524 mm. Therefore, the same trace width is set in *Ultiboard*, since the maximum current is roughly 30 mA. Other measures are left at default settings in both tools: drill hole at 0.8 mm for components, for header (power connector) 1.1 mm; pad diameter in *Fusion 360* is 1.26 mm – while in *Multisim* it is 1.5 mm; clearance (spacing between traces) in *Fusion 360* is 0.15 mm, while in *Multisim* it is 0.254 mm. Additionally, both tools automatically determine the hole diameter according to the component PCB model. The size of the board is then reduced to a size sufficient for the assembly. Considering the components used, the size is 29 x 26 mm in both tools. Next, the components are connected using the *Auto-routing* option, generally aiming for short and direct traces. The final PCB layout is shown in Fig. 5 and Fig. 6. Three-dimensional (3D) PCB layout is presented in Fig. 7. The next step, in both tools, is putting the '*polygonal pour*'. *Polygon pour* is a technique used to create large areas of copper on the board. These areas can be solid or hatched and are used for various purposes, e.g., large current distribution, electromagnetic shielding, connection, for heat dissipation. In this PCB is connected to GND terminal.

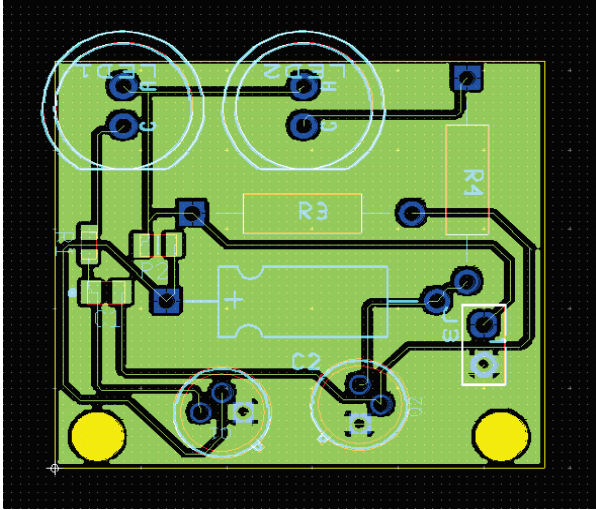


Figure 5. PCB layout in *Ultiboard*

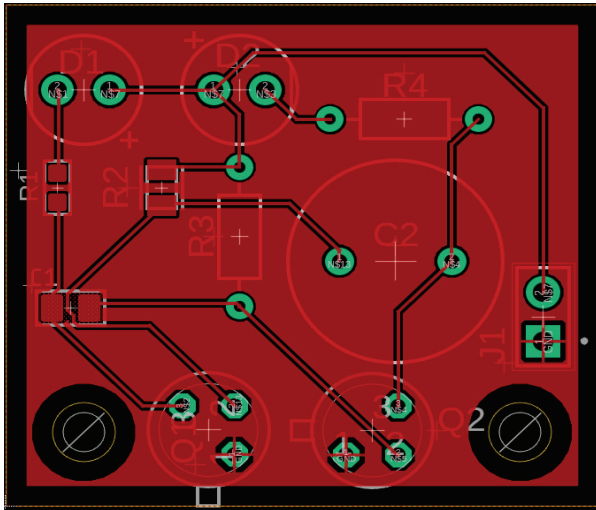


Figure 6. PCB layout in *Fusion 360*

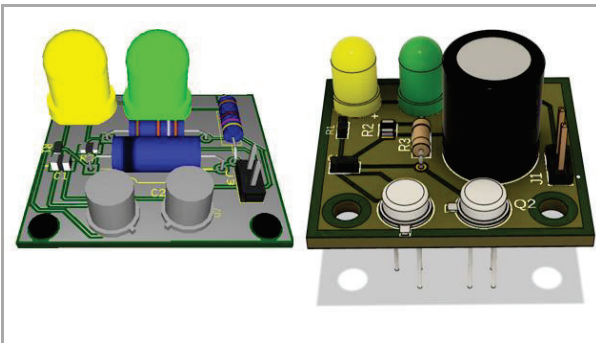


Figure 7. Three-dimensional PCB layout in *Ultiboard* and *Fusion 360* (right side)

The following step in design process is *Design Rule Check* (DRC). This process is used to verify that a PCB layout adheres to a set of rules or constraints. Some key aspects of DRC include: Spacing constraints (verifies and ensures adequate spacing between traces and components to meet maximum current requirements); Clearance rules (ensures sufficient clearance between traces); Component placement (verifies that components are placed correctly and have enough space around for proper soldering); and Power and Ground trace width (ensures that power and ground traces are wide enough to handle the required

current without overheating or voltage). In *Multisim*, DRC is performed automatically during component placement on the board. In *Fusion 360*, DRC have to be started after the connection of conductors on the board is completed. Since no errors were found, both tools report 'No errors', indicating that board is designed correctly.

Finally, based on the obtained results, it can be concluded that both tools are suitable and intuitive for PCB design in the mentioned phase.

V. PCB ANALYSIS

Analyzing the behavior of entire assembly under real-world conditions/circumstances, such as temperature fluctuations, components cooling, mechanical stress, etc., can provide insight into assembly's durability, component aging due to heating and mechanical resistance throughout product lifecycle. Simulations carried out are *Cooling* and *Stress*. Unlike *Fusion 360*, aforementioned simulations cannot be carried out in *Multisim/Ultiboard*.

Stress simulation in *Fusion 360*, provides results for *Displacement*, *Reaction force*, *Strain* and *Contact Pressure/Force*. For such simulations on entire assembly, the value of each component masses (collected from component datasheet) has to be entered along with gravity force. The default board material is FR4 (Flame Retardant 4), usually used for PCB. Obtained results for stress are shown in Fig. 8, where the gravity force is represented by a brown arrow, indicating the global effect of gravity on entire PCB. The simulation presents areas on the PCB with highest stress. Green areas with arrows indicate stress distribution in the board due to its weight. In the green areas, stress ranges between 7.54 and 8.51 MPa near the screw holes on a board.



Figure 8. PCB stress simulation in *Fusion 360*

The results for *Strain* on the board are shown in Fig. 9.



Figure 9. PCB strain simulation in *Fusion 360*

An *Equivalent strain* simulation was chosen, which includes both linear and angular deformations as normal and shear stress occur. The *Equivalent strain* is simulated due to simultaneous interaction of bending and shear stresses. Simulation presents areas of strain on the assembly, so it indicates no significant strain. In the green areas, strain is approximately 0.12 %, while on yellow

points in the board hole is 0.17 %.

The results for *Reaction force* in the board, which occurs in response to weight or stresses, are shown in Fig. 10 and indicate no significant force. The probe has to be set for indicating a numerical value at particular point. Placed probe indicates 0.26 N as the maximum reaction force in board (marked in green).

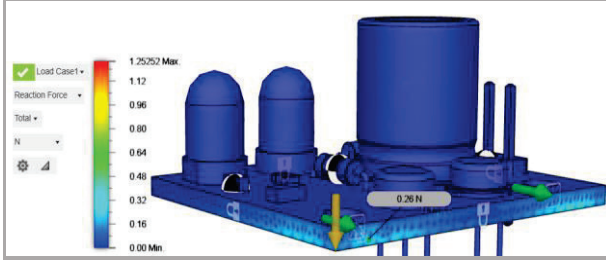


Figure 10. Reaction force in the board

Cooling simulation provides results for Component temperature, Maximum Thresholds, Air velocity, Air temperature, Air Velocity/Component temperature, Air temperature/Component temperature. For this analysis, power dissipation of each component has to be entered in *Fusion 360*. The ambient temperature is set to 27 °C.

Maximum Thresholds indicates the maximum allowed or possible values for a particular temperature on component with highest temperature value. The results are shown on Fig. 11. Along with maximum rated temperature, the component operating temperature is noted for components with highest temperature value.

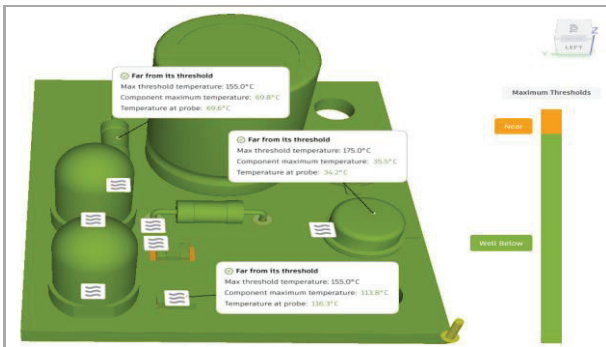


Figure 11. Maximum Thresholds for temperature

Component temperature simulation displays *steady-state* component temperature. This is the temperature that the component reaches after the thermal processes have stabilized and there are no further changes in temperature over time. Thus, the highest temperature is detected on R1: $T = 116.3\text{ }^{\circ}\text{C}$, (SMD housing; maximum rated temperature is $155\text{ }^{\circ}\text{C}$), while the temperature on R4 is $T = 69.6\text{ }^{\circ}\text{C}$. Due to elevated operating temperature, the component aging is accelerated as well as the aging of the board, which shortens assembly's lifecycle. In such situation, engineer/designer can redesign the PCB by adding thermal vias, or replace the SMD component with a radial one (which often has better heat dissipation due to its larger surface), or replace the existing component with one that has higher power dissipation (e.g., instead of existing 1/4 W component, use a 1/2 W component or 1 W component).

The results for *Air Velocity / Component temperature* are shown in Fig. 12. Since the assembly is intended to

operate in a closed room without draught, the air flow is set to $v = 0\text{ m/s}$. Due to temperatures of some components

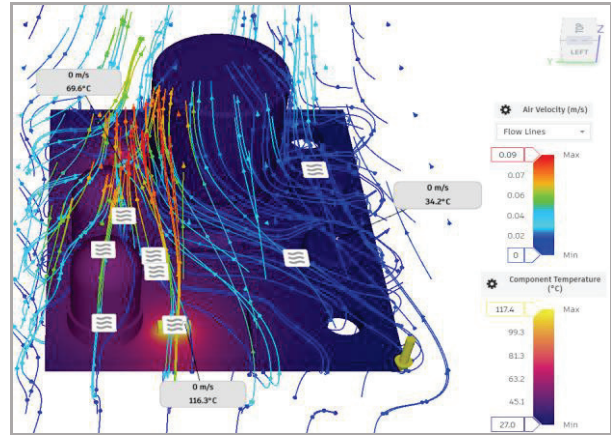


Figure 12. Results for Air Velocity/Component simulation

are significantly higher than the ambient temperature, air flow occurs, shown by red, orange and yellow flow lines. It lines indicates that air velocity around heated components is 0.06 – 0.09 m/s.

Despite that aforementioned analysis cannot be performed in *Multisim*, it does enable *Temperature Sweep* analysis, which allows simulation of a circuit operation at various temperatures. This analysis can be combined with other analyses, such as *DC Operating Point Analysis*, to examine changes in voltages and currents at various temperatures. Some of the obtained results are presented in Fig. 13 in tabular form (*Multisim* offers a choice between displaying the results in tabular form or graph). For clearer presentation, differential voltage probes are additionally placed to measure the voltage drop across the component (resistors) R1 and R4. The results show that as the temperature increases from $0\text{ }^{\circ}\text{C}$ to $120\text{ }^{\circ}\text{C}$, the currents

MIPRO2025 Temperature Sweep		
	Variable, Temperature setting	Operating point value
1	V(PR 1), Temperature=0	7.124
2	V(PR 2), Temperature=0	6.859
3	I(R 1), Temperature=0	15.158 m
4	P(R 1), Temperature=0	107.992 m
5	I(R 4), Temperature=0	14.593 m
6	P(R 4), Temperature=0	100.085 m
7	P(V 1), Temperature=0	-271.567 m
8	I(V 1), Temperature=0	-30.174 m
9	V(PR 1), Temperature=40	7.026
10	V(PR 2), Temperature=40	6.723
11	I(R 1), Temperature=40	14.949 m
12	P(R 1), Temperature=40	105.032 m
13	I(R 4), Temperature=40	14.303 m
14	P(R 4), Temperature=40	96.153 m
15	P(V 1), Temperature=40	-267.110 m
16	I(V 1), Temperature=40	-29.679 m
17	V(PR 1), Temperature=80	6.930
18	V(PR 2), Temperature=80	6.589
19	I(R 1), Temperature=80	14.744 m
20	P(R 1), Temperature=80	102.167 m
21	I(R 4), Temperature=80	14.018 m
22	P(R 4), Temperature=80	92.359 m
23	P(V 1), Temperature=80	-262.727 m
24	I(V 1), Temperature=80	-29.192 m
25	V(PR 1), Temperature=120	6.835
26	V(PR 2), Temperature=120	6.456
27	I(R 1), Temperature=120	14.542 m
28	P(R 1), Temperature=120	99.390 m
29	I(R 4), Temperature=120	13.737 m
30	P(R 4), Temperature=120	88.694 m
31	P(V 1), Temperature=120	-258.415 m
32	I(V 1), Temperature=120	-28.713 m

Figure 13. Temperature Sweep analysis results in *Multisim*

and voltages on the components change, and consequently part of the power is dissipated as heat. For example, current $I(R1)$ at 0 °C is 15.158 mA with a voltage drop $V(PR1)$ of 7.124 V; and at 120 °C, current $I(R1)$ is 14.542 mA, and $V(PR1)$ decreases to 6.835 V.

Finally, it can be concluded that advantage of *Multisim* lies in wider circuit analysis options as well as the ability to display results in both tabular and graph form, while the advantage of *Fusion 360* lies in its graphical display of phenomena and option to simulate assembly in the mechanical and cooling fields.

VI. DISCUSSION AND CONCLUSION

The objective of this paper was to compare the PCB design capabilities of the CAD tool *Fusion 360* with an already used EDA tool *NI Multisim/Ultiboard* within the course ‘*Circuit Analysis with EDA*’ at the graduate study of Electrical Engineering at the Zagreb University of Applied Sciences. The general conclusion is that CAD *Fusion 360*, along with EDA *Multisim/Ultiboard* is suitable for PCB design with some described limitations, but also with the additional possibilities.

Based on the obtained results, it is evident that the CAD tool cannot fully replace the EDA tool in observed domain. However, it is important to highlight that the CAD tool offers certain analyses that the EDA tool cannot perform. Consequently, both tools play a crucial role in the educational process, allowing students to gain insights into the behavior of the entire assembly under real-world conditions, such as temperature fluctuations, cooling, mechanical stress, etc., which determine assembly’s durability and component aging due to heating. The analyses and simulations performed, along with the corresponding phenomena displayed graphically and numerically, can contribute to an easier understanding of these phenomena. Furthermore, it enables a holistic approach to the assembly (and generally to engineering tasks) - not only from the aspect of functionality but also from the analysis of performance under certain conditions/circumstances. Therefore, the usability of these tools for educational purposes, extends beyond the scope of the course ‘*Circuit Analysis with EDA*’. It represents an opportunity to supplement and expand student’s knowledge. This is particularly relevant to courses such as *Engineering Mechanics* as well as *Materials and Manufacturing Processes*, where, in addition to ‘*manual*’ stress and thermal calculations, students can compare these results with those obtained using the CAD tool. An additional benefit of using both software tools in the educational process is their applicability in laboratory exercises across several courses, along with measurements with real equipment (e.g., *Analog Circuits*, *Digital Circuits*, *Electronic Components*). This approach allows students to understand the specifics of real measurements and measuring equipment while performing the same tasks in software tool through simulations. Moreover, the use of CAD enhances already acquired skills in 2D drawing and 3D modeling within a familiar and proven user interface (used on courses *Technical documentation* and *3D modelling and 3D printing*).

From a student’s perspective, CAD offers a simple and intuitive interface: the toolbar is organized to follow the workflow, listing the next steps from schematic capture to

analysis of the entire assembly on the PCB. Namely, this paper was initiated and co-created by the student (one of the co-authors) as part of a student’s research for an upcoming master’s thesis. Additionally, the perception is that *Multisim* has a more ‘*conservative*’ toolbars layout, which may seem less intuitive to the student compared to *Fusion 360*.

From a teaching perspective, using one software tool for wide range of tasks is a more time-efficient way of transferring knowledge. This approach eliminates the need for students to learn a new user interface for several tools, and it also contributes to avoid possible errors in manually data transferring between tools.

Future steps could include:

- collecting feedback from a larger number of students after using both tools in PCB design, as well as in the mentioned courses for stress and thermal simulations,
- analyzing the impact of combining software tool with real measurement equipment during laboratory exercises in the mentioned courses on the overall educational process, alongside with collecting feedback from students,
- comparing capabilities of CAD and EDA tools in designing more complex circuits, including integrated circuits and those operating at high frequencies with active cooling simulation.

Both tools are commercial. Unlike that, the *Fusion 360* is free of charge for university and students with their educational license. *Multisim* is commercial for university and for students offers educational license free of charge.

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